

Institute of Actuaries of India

Subject ST5 – Finance and Investment A

May 2013 Examination

INDICATIVE SOLUTIONS

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1 :-

- (i) A warrant is an option issued by a company. The holder has the right to purchase shares by paying a specified price at specified times in the future.

Reasons for issue of warrants

- Usually issued to investors who subscribe to bonds as a deal sweetener/equity kicker. Warrant provides an opportunity to the bond investor to participate in the upside in stock if the company does well. The company benefits because they can negotiate a lower interest rate with the investors.
- Promoters of a company may use warrants as a route to increase their ownership. Since the promoters need to pay up only 25% margin up front (in India) they can enjoy the benefits of stock ownership without paying the full amount up front.

(3)

(ii)

Warrant	Call option
Issued by a company to investors and the warrant is guaranteed by the issuing company	Is a contract between 2 investors and is independent of the company and the guarantee to honour the trade may be provided by the exchange
Customizable – terms decided by issuer	Standardized product – terms decided by the exchange
On exercise leads to equity dilution. New funds are made available to the company	No equity dilution on exercise. No fund is available to the company. One investor gains and another investor loses in this zero sum game
Longer time to expiry (>18 months)	Shorter time to expiry (<3 months)

(4)

- (iii) CFDH Ltd. – A renowned housing finance company had issued warrants to investors. The warrants have a strike price of 600 and are listed on National Stock Exchange. The expiry date is 23rd Aug 2012. To exercise a warrant the holder needs to fill an application form and send it to CFDH Ltd. along with a Demand Draft for Rs. 600/warrant and CFDH will issue one share for every warrant. The whole process will take about 15 days. On 20th of July 2012 a hedge fund manager sees the price of the warrants and stock over the last few trading days. The following is table gives you the price of the warrant and the underlying.

Date	Warrant Price	Stock Price
19-Jul-12	81.66	693.55
18-Jul-12	74.98	686.75
17-Jul-12	72.53	681.60
16-Jul-12	70.66	678.60
13-Jul-12	68.27	674.95
12-Jul-12	67.99	674.45
11-Jul-12	69.90	678.30
10-Jul-12	71.48	682.55
09-Jul-12	68.79	675.45
06-Jul-12	75.82	683.65

At first glance the hedge fund manager believes that there is an arbitrage opportunity here. Why?

Besides CFDH stock being liquid, fairly liquid futures and options on CFDH underlying stock are also available, How can an arbitrage trade be structured to take advantage of this? *The lot size for warrant is 1850 and that of futures and options is 500.*

Reasons to suspect an arbitrage opportunity at first glance

- The minimum price at which the warrant should trade is its intrinsic value (Stock price minus exercise price of 600). On most of the days the warrant price is less than the intrinsic value and since there is time remaining for expiry there should be a time value component also which means there is an arbitrage opportunity.
- How the trade can be structured
 - The fund manager can go long on CFDH warrants and short on CFDH futures.
 - If he believes that this is a temporary pricing anomaly that will get corrected he can determine the hedge ratio on the basis of the delta of the warrant.
 - Or he can simply take a fully hedged position convert the warrant into stock and unwind the position upon the shares getting credited to his account.

(5)

iv)

- Is there a dividend that is going to be declared by CFDH with ex-dividend date between today and date of exercise? This possibly explains the discount.
- Conversion of warrant into stock requires a substantial investment albeit for a small amount of time. The acquisition cost per lot of the warrant at say a price of 81.66 is $1850 * 81.66 = 1,51,071$. But to convert this warrant into stock one needs to invest additional $1850 * 600 = 11,10,000$ which is a pretty substantial investment. If the discount to intrinsic value continues to remain the arbitrage transaction will materialize only on conversion. So is there enough liquidity in the fund?
- If the fund manager has to convert he needs to consider the opportunity cost of 600 per warrant for a period of 15 days till the stock is credited to his account from the company. Say at 10% this works out to about 2.5 per warrant, the present value of which needs to be knocked off from the price of the warrant.
- If the hedging is done using July 2012 futures then there will be a rollover risk when the position is carried forward to August and if required again in September. If the futures are at a discount for some reason there will be losses on rollover.

(8)

[Total 20 Marks]

Solution 2 :-

(i)

- Cut in interest rates makes personal loans (like mortgage loans) cheaper and increases disposable income of individuals which will increase consumption.
- Lower interest discourages savings and encourages consumption.
- Lower interest rates reduce the cost of financing for projects. Hurdle rates are lower and more projects become viable boosting the investment demand. Boosting investment often has profound influence on the GDP due to its multiplier effect.
- More investment often leads to higher job creation and increased standard of living. This can boost consumption.
- Lower interest rates reduce the demand for rupee leading to depreciation of currency making it more competitive for exports. Cut in interest rates boost exports because of the impact on currency. Similarly, imports become costlier and may result in overall reduction in imports.

(4)

(ii)

- A current account deficit happens when a country imports more goods and services than it exports. A cut in interest rates leads to a lower demand for domestic currency and leads to depreciation of the domestic currency. Since the country is a net importer and many imported items like petroleum products are necessary goods, a depreciated currency can further accentuate the problem of current account deficit.
- Inflation refers to general increase in level of prices for goods and services. A cut in interest rate encourages consumption and leads to increased demand for goods and services and hence tend to increase inflationary pressure.
- As the keeper of monetary policy, RBI tries to balance various economic parameters like economic growth, inflation, current account deficit etc. So if current account deficit and inflation levels present currently are not within the acceptable limits of RBI, it may not be too keen to cut rates even if economy is slowing down. (5)

(iii)

- All the gold that is purchased by India is imported. Purchase of gold is treated as consumption demand. So the purchase of gold by Indians leads to significant increase in value of total imports and worsens the current account deficit.
- The government reckons that the money being spent to purchase gold is money not being put to any productive use. Since they believe this is an unproductive expenditure which has the added pitfall of worsening the current account deficit they are mulling measures to curb the demand for gold. (2)

(iv)

- **Advantage:**
 - This instrument is essentially a derivative that depends on the price of gold. So investors who buy gold for investment purposes can use this.
 - They are saved from the hassle of storing physical gold (gold has negative cost of carry) but get all the benefits of the increase in price of gold that they foresee.
 - Since this is not backed by physical gold, purchase of this note will not lead to worsening of current account deficit and hence may get a more friendly treatment from the government

- **Disadvantage:** The seller of the note will be having an unhedged position and increase in price of gold will be detrimental to the seller as it is not backed by physical gold. So the buyer is exposed to credit risk in the event of default by the seller. (3)

(v)

- This treatment can of course bring down the reported current account deficit number. But irrespective of the way in which it is treated the outflow of money to purchase gold continues to happen and the associated problems continue to remain. (1)

[Total 15 Marks]

Solution 3 :-

(i)

- Credit rating agencies publish default loss surveys for fixed interest securities. Develop a matrix of aggregate probability of default for a given term and a given initial credit rating.
- Allocate initial ratings to each security in the portfolio. Use the current rating if rated and an approximate rating if not rated. e.g. sovereign rating down say 2 notches dependent on size and known corporate factors.
- Use the probability of default matrix term structure to derive the probability of a given “n” year security defaulting in each of year’s 1 to “n” inclusive.
- Adopt a table of default recovery rates reflecting the security of given security types from available experience.
- Allocate a recovery rate to each security. The default loss of a given security is then determined from its probability of default and its recovery rate.

(5)

(ii)

- Adopt a correlation matrix that provides a correlation coefficient for any two different securities.
- The correlation coefficient for any two securities can be the sum of assumed correlation coefficients between, say, geographic regions, countries within regions, and same or similar industry.

(2)

[Total 7 Marks]

Solution 4 :-**(i)**

- To ensure that financial instruments are housed under a proper system that permits investment for proper purposes with proper authority. **(1)**

(ii)

- A custodian trustee not only has physical custody of documents but also holds the legal title to the trust assets.
- Only specified categories of corporation, known as trust corporations, are empowered to act as custodian trustees. **(2)**

(iii)

- Cost
- Transfer issues
- Range of services offered. Range of services required e.g. income collection, cash management.
- How the custodian operates globally on a country by country basis. E.g. through sub-contracts with local custodians or directly with the Central Securities Depositories.
- Corporate structure e.g. capital adequacy, approach to operational risk/fraud, demonstrable competence
- credit rating
- regulatory approval
- size, administrative systems and integration to the investment manager potential to streamline manager's own costs of dealing with the custodian, number.

(4)**[Total 7 Marks]****Solution 5 :-****(i)**

- Each capital index has its particular purpose and use
- Each index will be constructed using certain eligibility criteria to determine the constituent companies from time to time
- Usually market capitalisations of the constituent companies are used as weights so typical portfolio will be weighted toward bigger companies
- Index typically start at fixed value e.g. 100 or 1000

- It is common practice to restrict the weights reflect the level of free float of shares available for purchase
- It is calculated periodically and with the advanced computer power it is calculated real time.

The formula can be used is as follows.

$$I(t) = K \frac{\sum_i w_i \frac{P_{i,t}}{P_{i,0}}}{\sum_i w_i}$$

- $I(t)$ is the capital index at time t
- $P_{i,t}$ is the price of the i th constituent at time t
- $P_{i,0}$ is the price of the i th constituent at time 0
- W_i is the weight applied to the i th constituent
- K is a constant related to the starting value of the index at time 0

(7)

(ii) The chain-linking process is set up so that:

- The index reflects the new market capitalisation of the constituents
- The index value is not disturbed by the change in capital structure of its constituents.

This is appropriate because if the aim of the index is to reflect the investment performance of its constituents, then the index value should change only in response to that investment performance and not in response to injections or withdrawals of funds into or out of the market itself.

Circumstances in which chain-linking would be required include:

- A rights issue or a share buy-back by a constituent company
- A new issue of a share in the sector covered by the index – e.g. newly-formed company, privatisation, demutualisation
- A merger, takeover or break-up involving a constituent company or companies
- A change in the constituent companies in an index, resulting from a change in relative market capitalisations due to share price movements. For example, the share price of Company P increases so that its overall market capitalisation now exceeds that of Company Q, which was formerly the 100th largest company. Thus, P replaces Q in the index of the largest 100 companies.

(4)

[Total 11 Marks]

Solution 6 :-**(i)**

- Prescriptive: Regulations with detailed rules setting out what may or may not be done.
- Freedom of action: Regulations which involve freedom of action but with rules on publicity so that third parties are fully informed about the providers of financial services.
- Outcome-based: Regulatory regime that allows freedom of action but prescribe the outcomes that will be tolerated.

(ii)

Reasons for having this regulation

- A form of protection to current promoters of the company from hostile takeover
- In case the promoter/major shareholder sells out to a buyer this provision gives a partial exit opportunity to other minority investors who may not be comfortable with the new buyer.

Shortcomings of the regulation

- Taking a sizeable position pretty difficult for big investors. Even if one takes a 15% stake they are as worse off as other minority investors.
- While the promoter/major shareholder can exit in one go usually at premium to current market price) the minority shareholders usually get only a partial exit if they are not comfortable with the buyer.

(iii)

- Since the threshold for open offer is increased to 25% the buyer can take a pretty large influential position even before the open offer is triggered. For example a 26% ownership is all that is needed to block board resolutions.
- Since the open offer is for all the remaining shares all the investors have the option to exit fully if they are not comfortable with the buyer. All shareholders are hence treated equally. **(1)**

(iv)

- Giving open offer to all the shareholders would mean that the cost of acquisition goes up substantially and this can scuttle the possibility for mergers and acquisitions. The industry prefers an economic environment where mergers and acquisitions can be done pretty easily. The acquirer would always prefer an option to take a controlling stake with the least possible investment.

- Since the buyer needs to give an open offer upon reaching 25% gathering another 26% from the market will take his stake to 51% which will give the buyer control over the company. The minority shareholders can get only a partial exit but is still much higher than what he got earlier.

(2)

[Total 10 Marks]**Solution 7 :-**

(i)

- Theoretically shareholders should reduce the company's target return on equity (ROE) to reflect the excess cash position. In practice this is not done as shareholders assume that the excess cash is a temporary position rather than a long-term strategy.
- Cash invested is reducing the company's ROE. It needs to be invested at or above the hurdle ROE or given back to shareholders.

(2)

(ii)

- Company's announced targeted ROE
- Company's actual ROE during each of the last three years
- Industry sector ROE for the capital project if the capital project is in a different industry sector
- Prospective earnings before interest and tax (EBIT) for the company net of the cash and prospective EBIT for the capital investment if the capital project is in a different industry sector

(4)

(iii)

Reasons for market valuing the cash at a discount :-

- Agency cost of equity – Costs arising from the fact that managers may maximise their own value rather than that of shareholders. This happens when the incentive structure of the management is not aligned with that of shareholders which leads to shareholders not fully trusting the management.
- Risk associated with a new business/project is usually much higher than that associated with running an extant business. The lower probability of success is reflected in the discount.

- Cost of equity is much higher than the return that cash can earn. Any delay in deployment of cash if foreseen by shareholders will lead to them valuing the cash at a discount. (4)

Solution 8 :-

(i)

- Money weighted rate of return (MWRR) is calculated by finding the rate of return that will set the present values of all cash flows and terminal values equal to the value of the initial investment. The money-weighted rate of return is equivalent to the internal rate of return (IRR). The standard formula for calculating MWRR is

$$V_0(1+i)^T + \sum_t C_t(1+i)^{T-t} = V_T$$

Time weighted rate of return (TWRR) is calculated by

- Assessing the fund value at each time there is a cash flow in or out
- Calculating the return achieved for each period between cash flows
- Linking these returns together to give the time-weighted rate of return. The formula for TWRR is

$$(1+i)^T = \frac{V_{t_1}}{V_0} \times \frac{V_{t_2}}{V_{t_1} + C_{t_1}} \times \dots \times \frac{V_T}{V_{t_n} + C_{t_n}}$$

where:

- V_0 is the market value of the fund at the beginning of the period
 - V_T is the market value of the fund at the end of the period of T
 - C_{t_r} is the net cash inflow, excluding investment proceeds, at time t_r
 - V_{t_r} is the market value of the fund at time t_r just before the cashflow
 - i is the time-weighted rate of return.
- TWRR is preferable as a basis for comparing different investment managers because the timing and size of cash flows will not distort the rates calculated. TWRR can be considered as a better indicator of fund manager's performance as cash inflow and outflow from the fund is not under the manager's control. (5)

(ii)

- MWRR can be calculate by solving the following equation for 'i'

$$102(1+i) + 12(1+i)^{0.75} - 30(1+i)^{0.5} + 100(1+i)^{0.25} = 190$$

Using the approximation $(1+i)^n = 1+ni$ the above equation can be solved for i and this works out to about 5%.

TWRR can be calculated as follows

Date	MV of equity (INR Cr)	Cash inflow/outflow	MV after Cash Flow	Return during the period
31-Dec-07	102	0	102	
31-Mar-08	72	12	84	-29.4%
30-Jun-08	75	-30	45	-10.7%
30-Sep-08	50	100	150	11.1%
31-Dec-08	190	0	190	26.7%

$$TWRR = \{(1-0.294)(1-0.107)(1+0.111)(1+0.267)\}-1$$

$$TWRR = -11.3\%$$

- The performance of NIFTY during the period was -10.4%. The fund manager's claim of superior performance is probably by looking at MWRR on which basis there is a significant outperformance over the NIFTY. But this can be attributed to sheer luck as the fund manager happened to receive a large cash inflow just before the market boomed in the quarter ending Dec 2008. Looking at TWRR however indicates that the fund manager performance was actually worse than the index. (8)

(iii)

- Sharpe ratio is a method to measure risk adjusted returns of a portfolio. It quantifies the excess returns generated per unit of risk taken. The ratio is calculated as follows:

$$S = \frac{R_p - r}{\sigma_p}$$

Where

- R_p is the return on the portfolio
- r is the risk free rate of return over the period
- σ_p is the standard deviation of portfolio returns over the period
- Sharpe ratio helps us to compare the performance of fund managers on a risk adjusted basis. When 2 fund managers are compared the fund manager with a higher Sharpe Ratio is preferred. (3)

(iv)

Fund	Sharpe Ratio
Fund 1	$\frac{-11.3\% - 7\%}{30\%} = -0.61$
Fund 2	$\frac{-11\% - 7\%}{15\%} = -1.2$

- Comment: While Fund 1 has a higher Sharpe ratio the result is intuitively incorrect as Fund 2 has generated nearly the same return as fund 1 with a much lower volatility. The applicability of Sharpe ratio as a comparison tool when excess returns are negative is questionable.

(4)

[Total 20 Marks]
